Ligjerata 3b

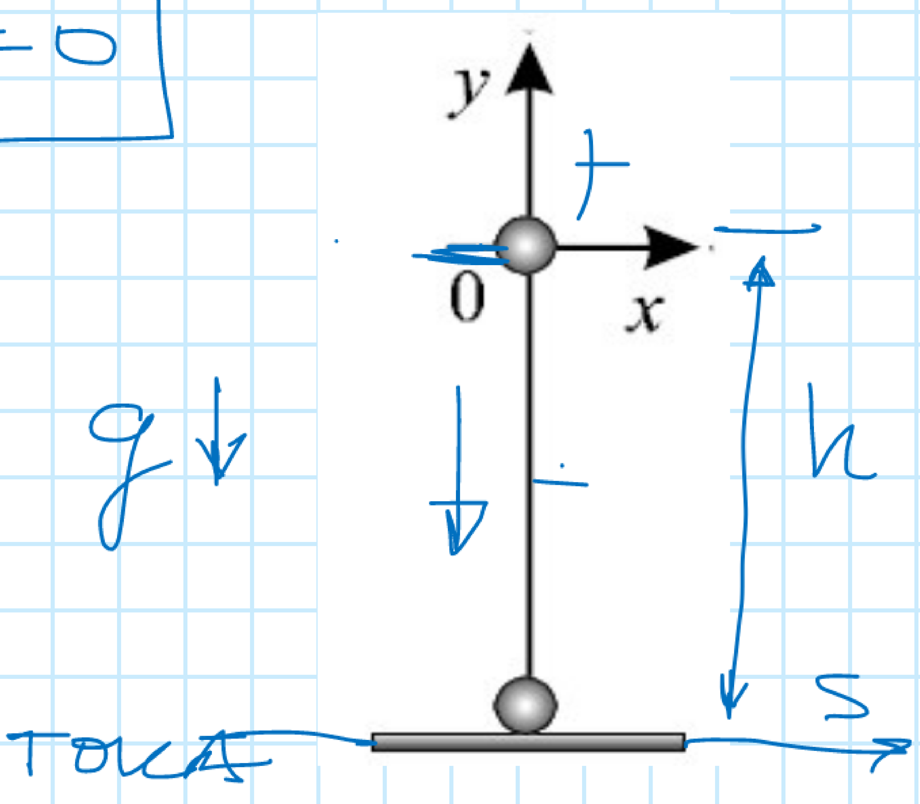
LËVIZJA E TRUPAVE NË FUSHËN GRAVITACIONALE

Rënia e lirë

$\boxed{h \ll R_T \quad F_f = 0}$ w

$\vec{a} = \vec{g}$ $\boxed{v_0 = 0}$

$g = 9.81\,\frac{m}{s^2}$



$g \downarrow$

$a = g = \frac{dv}{dt} \;/\cdot dt$

$dv = g\,dt \;/\int$

$\int dv = g\int dt$

$v = g\,t + C$

$t = 0,\; v = 0,\; C = 0$

$\boxed{v = gt}$ sh. funksion i kohës

$v = \frac{ds}{dt} \;/\,dt$

$ds = v\,dt = gt\,dt \;/\int$

$\int ds = g\int t\,dt$

$s = g\frac{t^2}{2} + C'$

$t = 0,\; S = 0,\; C' = 0$

$S = g\frac{t^2}{2},\; s = h$

$h = g\frac{t^2}{2} \rightarrow \frac{2h}{g} = t^2 \;/\sqrt{\ }$

$t = \sqrt{\frac{2h}{g}}\;;\; v = gt$

$v = g\sqrt{\frac{2h}{g}} = \sqrt{\frac{g^2 \cdot 2h}{g}}$

$\boxed{v = \sqrt{2gh}}$ sh. si funksion i lartësisë.

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$s = v_0 t + \frac{1}{2} a t^2 \quad (*)$

$v = v_0 + at \quad (**)$

$s = h \quad a = g$

$v_0 = 0$

$h = 0 \cdot t + \frac{1}{2} g t^2$

$\boxed{h = \frac{1}{2} g t^2}$

$\boxed{v = gt}$

$t = \sqrt{\frac{2h}{g}}\;;\; v = gt$

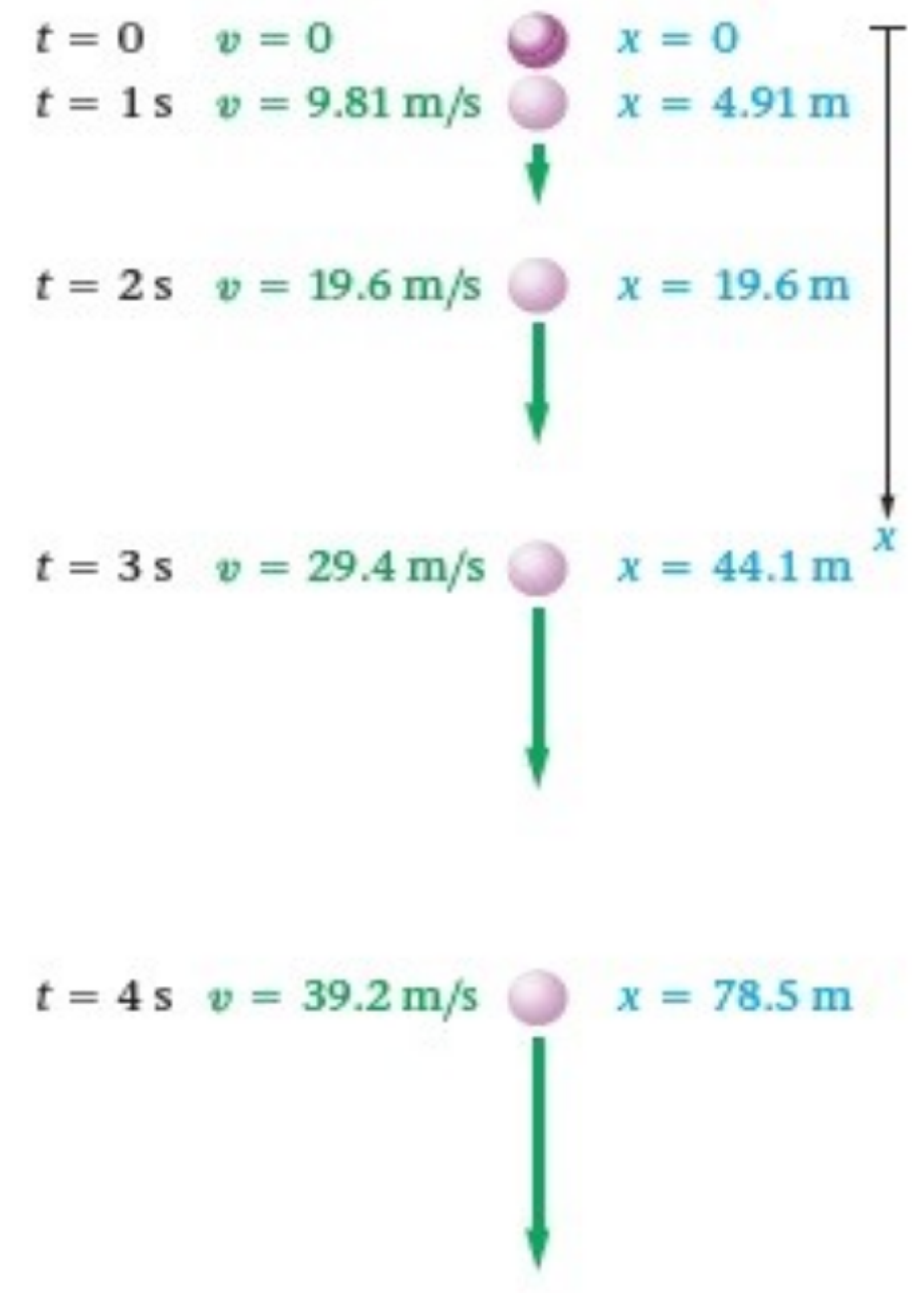
$v = g\sqrt{\frac{2h}{g}} = \sqrt{\frac{g^2 \cdot 2h}{g}}$

$\boxed{v = \sqrt{2gh}}$ sh. si funksion i lartësisë.

$v[=]\,\frac{m}{s}$

$g[=]\,\frac{m}{s^2}$

$h[=]\,m$



Te vetrtetohen vlerat numerike per:
x=h=?
v=?

D. sh.